

Introducing the Open Source IaaS Platform

In its most basic form, Infrastructure as a Service (IaaS) includes computers – physical or virtual machines – and other computer related services. IaaS distances end users from the task and expense of setting up computer systems and maintaining them, so that they can concentrate on their special fields of interest.

IaaS, which stands for Infrastructure as a Service, is a cloud computing platform that provides computing services over the Internet.

What is IaaS?

In an IaaS model, the provider takes care of hosting servers, the storage, applications and any other required hardware, all on the provider's premises. This infrastructure is highly scalable and provides business resiliency. Examples of IaaS service providers are Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), OpenStack, Eucalyptus, CloudStack, etc.

OpenStack

OpenStack is open source IaaS software stack that is mainly used for deploying or managing large groups of virtual private servers in a data centre.

The goals of the OpenStack initiative are to support interoperability between cloud services and to allow businesses to build cloud services within their own data centres.

It has a modular architecture with the following components:

- OpenStack Identity (Keystone)
- OpenStack Dashboard (Horizon)
- OpenStack Compute (Nova) including the use of multiple hypervisor drivers
- OpenStack Object Storage (Swift) or another object storage solution
- OpenStack Block Storage (Cinder)
- OpenStack Image Service (Glance)
- OpenStack Networking (Neutron) or Legacy Networking (Nova-network)

NASA and Rackspace jointly started this project with

a few components, and it has now grown to a mammoth scale. OpenStack enables users to create and offer cloud computing services. It is entirely a community-driven project, and there is a new release every six months. OpenStack consists of many projects that constitute various functionalities



Figure 1: OpenStack



like compute, storage, network, security/authentication, automation and user interface. The current stable release is called Juno. The latest release also includes data processing capabilities. The major components of OpenStack in the latest release are: Nova, which is responsible for compute; Neutron, which is responsible for networking; Swift, for providing object storage; and Cinder, which takes care of providing block storage to virtual instances. Keystone ensures identity/authentication, Glance provides storage space to hold images and VM templates, and Horizon provides a decent dashboard to manage all resources for admins and users. Ceilometer provides the resource utilisation stats and this, in turn, is used in billing. The project Heat provides automation/orchestration, Trove enables database services, and project Sahara provides the capability for data processing.

Apart from these components, a few more are in development – the Ironi project that will provide the capability to do bare-metal provisioning, Manila which will provide shared file systems, and a few more. Refer to openstack.org for more information on these and other projects.

Eucalyptus

Eucalyptus is another open source project for providing cloud infrastructure as a service. Eucalyptus is an acronym for 'Elastic Utility Computing Architecture for Linking Your Programs